

Program : Diploma in Polymer Technology	
Course Code : 4097	Course Title: Ingredients Testing Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To provide expertise in the estimation of purity of compounding ingredients
- To impart skill in the determination of properties of carbon black from iodine number of black

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I	1, 2
Basic Knowledge in chemistry		Applied Chemistry	1, 2

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive level
CO1	Determine the Purity of compounding ingredients	24	Understanding
CO2	Determine the properties of carbon black.	15	Understanding
	Series Exam	6	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1		3		3			
CO2		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Determine the Purity of compounding ingredients		
M1.01	Determine the percentage purity of MBT	3	Understanding
M1.02	Determine the percentage purity of DPG	3	Understanding
M1.03	Determine the percentage purity of ZnO	3	Understanding
M1.04	Estimate the percentage purity of formalin.	3	Understanding
M1.05	Determine the moisture content of clay and silica	3	Understanding
M1.06	Determine the melting point of stearic acid,TMT.	3	Understanding
M1.07	Determine the acid number of stearic acid.	3	Understanding
M1.08	Determine the aniline point of processing oil.	3	Understanding
	Lab Exam I	3	
CO2	Determine the properties of carbon black		
M2.01	Determine the iodine number of HAF black	6	Understanding
M2.02	Determine the iodine number of GPF black	6	Understanding
M2.04	Determine the Bulk density of carbon black	3	Understanding
	Lab exam II	3	

Text / Reference:

T/R	Book Title/Author
R1	Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd, 1982 Edition)
R2	Rubber Technology -Maurice Morton (Publisher – Springer, 1999 Edition)
R3	Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)
R4	Training Manual - RTI, RRII, Kottayam
R5	Rubbery Materials and their compounds - J.A Brydson (Publisher –Elsevier Applied Science, London, 1988)

Online Resources:

Sl.No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	www.iri.net.in
4	www.sciencedirect.com